

SDR Monitoring — Pseudocode Example

pseudocode-specific skill

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Overview

High-level SDR monitoring pipeline: SoapySDR → IQ acquisition → Welch PSD → Plot.

Listing 1: High-level flow (pseudocode-general)

```
1. IMPORT LIBRARIES (SoapySDR, numpy, scipy.signal, matplotlib)
2. sdr = SETUP SDR DEVICE (driver="rtlsdr", sample_rate=2.4e6, center_freq=100e6, gain
   =40)
3. iq = ACQUIRE IQ STREAM (sdr, num_samples=65536)
4. f, psd = APPLY WELCH METHOD (iq, fs=2.4e6, nperseg=1024, noverlap=512)
5. PLOT frequency spectrum (f, psd, xlabel="Frequency (Hz)", ylabel="PSD (dB/Hz)")
6. SAVE figure ("output/sdr_welch.png")
7. LOOP forever:
   7.1 iq = ACQUIRE IQ STREAM (sdr, num_samples=65536)
   7.2 f, psd = APPLY WELCH METHOD (iq, fs=2.4e6, nperseg=1024, noverlap=512)
   7.3 UPDATE plot (f, psd)
   7.4 PAUSE (0.1s)
```

Detailed Pseudocode

Listing 2: SDR Monitoring — pseudocode-specific conventions

```
DECLARE SAMPLE_RATE AS FLOAT <- 2.4e6
DECLARE CENTER_FREQ AS FLOAT <- 100e6
DECLARE GAIN AS FLOAT <- 40
DECLARE NUM_SAMPLES AS INTEGER <- 65536
DECLARE NPERSEG AS INTEGER <- 1024
DECLARE NOVERLAP AS INTEGER <- 512

FUNCTION SetupSDRDevice(driver, sample_rate, center_freq, gain)
    DECLARE sdr AS DEVICE
    <- SoapySDR.Device(driver)
    <- sdr.setSampleRate(sample_rate)
    <- sdr.setFrequency(center_freq)
    <- sdr.setGain(gain)
    RETURN sdr
END FUNCTION

FUNCTION AcquireIQStream(sdr, num_samples)
    DECLARE iq AS ARRAY
    DECLARE buffer AS ARRAY of COMPLEX
    <- sdr.setupStream(SOAPY_RX)
    <- sdr.readStream(buffer, num_samples)
    <- buffer.astype(COMPLEX64)
    RETURN iq
END FUNCTION

FUNCTION ApplyWelchMethod(iq, fs, nperseg, noverlap)
    DECLARE f AS ARRAY
    DECLARE psd AS ARRAY
    (f, psd) <- scipy.signal.welch(iq, fs=fs,
                                   nperseg=nperseg, noverlap=noverlap)
    RETURN (f, psd)
END FUNCTION

FUNCTION PlotSpectrum(f, psd, title)
    DECLARE fig AS FIGURE
    <- matplotlib.pyplot.figure()
    <- matplotlib.pyplot.plot(f, psd)
    <- matplotlib.pyplot.xlabel("Frequency (Hz)")
    <- matplotlib.pyplot.ylabel("PSD (dB/Hz)")
    <- matplotlib.pyplot.title(title)
    <- matplotlib.pyplot.grid(TRUE)
    RETURN fig
END FUNCTION

FUNCTION SaveFigure(fig, path)
    <- fig.savefig(path)
END FUNCTION

FUNCTION ContinuousMonitoring(sdr)
    DECLARE f AS ARRAY
```

```

DECLARE psd AS ARRAY
DECLARE fig AS FIGURE
WHILE TRUE = TRUE
    iq <- AcquireIQStream(sdr, NUM_SAMPLES)
    (f, psd) <- ApplyWelchMethod(iq, SAMPLE_RATE,
                                NPERSEG, NOVERLAP)
    fig <- PlotSpectrum(f, psd, "SDR Monitoring")
    matplotlib.pyplot.pause(0.1)
END WHILE
END FUNCTION

--- Main ---
sdr <- SetupSDRDevice("rtlsdr", SAMPLE_RATE, CENTER_FREQ, GAIN)
iq <- AcquireIQStream(sdr, NUM_SAMPLES)
(f, psd) <- ApplyWelchMethod(iq, SAMPLE_RATE, NPERSEG, NOVERLAP)
fig <- PlotSpectrum(f, psd, "SDR Monitoring")
SaveFigure(fig, "output/sdr_welch.png")
ContinuousMonitoring(sdr)

```